THE 34 FACTORS & 8 BUSINESS CLUSTERS

Coripet RVM Location Predictor • Certified Factor Allocation for Production Stacking Ensemble *(LightGBM 90% + MultiTaskRVMNet Neural Network 10%)*. All weights normalized to exactly 100.00%.

1. Overall Cluster Weight Distribution (Line-by-Line A to H)

BUSINESS CLUSTER	FACTORS	NEURAL NET	LIGHTGBM	PRODUCTION FINAL	ROUNDED %
A. Store Physical Scale & Capacity	6	13.96%	8.05%	**8.64%**	**9%**
B. Municipal Demographics & Density	3	6.56%	8.74%	**8.52%**	**9%**
C. Territorial Wealth & Economics	2	4.40%	4.31%	**4.32%**	**4%**
D. Geographic Positioning	6	25.50%	34.61%	**33.70%**	**34%**
E. Spatial Network & Competitive Density	6	12.82%	6.72%	**7.33%**	**7%**
F. Hardware Technology	2	10.22%	8.28%	**8.47%**	**8%**
G. Retail Brand & Buying Alliance	2	12.61%	21.53%	**20.64%**	**21%**
H. Temporal & Seasonality Cycles	7	13.93%	7.76%	**8.38%**	**8%**
TOTAL (SUM OF ALL CLUSTERS)	**34**	**100.00%**	**100.00%**	**100.00%**	**100%**

2. Strategic Architectural Insights

• **Geographic Anchor (33.7%):** Italy exhibits pronounced regional variation in civic recycling velocity. The 107 Italian Provinces (Provincia) serve as the primary macro-anchor, capturing local waste tax (TARI) habits and regional voucher redemption behaviors.
• **Retail Brand Power (20.6%):** Commercial supermarket brands (Conad, Coop, Esselunga, Bennet) and buying consortiums represent the second largest driver, reflecting loyal weekly shoppers and promotion appeal.
• **Physical Store Scale (8.6%) & Hardware (8.5%):** Sales area (m²), checkout registers, and hardware model define the operational intake ceiling during peak customer dropoff hours.
• **Temporal Dynamics (8.4%):** Captures weekly rhythms (Saturday grocery peaks) and annual summer hydration ramp-ups.

Optional View: 7 Physical Store Clusters (A to G, excluding cyclical Temporal):
D. Geographic Positioning: 36.78% • **G. Retail Brand & Alliance:** 22.53% • **A. Store Physical Scale:** 9.43% • **F. Hardware Technology:** 9.24% • **B. Demographics & Density:** 9.30% • **E. Spatial & Competition:** 8.00% • **C. Territorial Wealth:** 4.72% • **Total = 100.00%**

3. Complete Catalog of All 34 Factors by Business Cluster

Comprehensive inventory of features, functional descriptions, and individual model weight contributions:

CLUSTER & FACTOR NAME	FUNCTIONAL DESCRIPTION / ENCODING	NN WT	LGBM WT	PROD WT
A. Store Physical Scale	*Sales floor, checkout capacity, retail format*	**13.96%**	**8.05%**	**8.64%**
• nielsen_tipo_negozio	Store format (LIS Proximity / SUP Supermarket / IPR Hyper)	4.10%	0.58%	0.93%
• nielsen_superficie_vendita_log	Sales Area (m²), log-transformed with median imputation	2.24%	3.16%	3.07%
• nielsen_numero_casse	Cash Register Checkout Throughput Count	2.04%	4.29%	4.07%
• nielsen_superficie_missing	Binary indicator for estimated vs verified sales surface	2.00%	0.02%	0.22%
• nielsen_casse_missing	Binary indicator for estimated vs verified checkouts	1.87%	0.00%	0.19%
• is_nielsen_matched	Database verification status in national grocery registry	1.72%	0.00%	0.17%
B. Demographics & Density	*Resident population, surface area, territory density*	**6.56%**	**8.74%**	**8.52%**
• population_log	Municipal resident population (ISTAT census, log-scale)	2.25%	2.62%	2.58%
• population_density_log	Population density per km² (urban vs rural trade zone)	2.18%	3.27%	3.16%
• surface_log	Municipal land area surface (km²)	2.13%	2.85%	2.78%
C. Territorial Wealth	*Macroeconomic purchasing power & taxable income*	**4.40%**	**4.31%**	**4.32%**
• PIL_per_capita	Per-capita taxable income / GDP indicator (€/person)	2.24%	2.06%	2.08%
• PIL_log	Aggregate municipal GDP economic scale (log-transformed)	2.16%	2.25%	2.24%
D. Geographic Positioning	*Province, region, postal routing, and GPS coordinates*	**25.50%**	**34.61%**	**33.70%**
• Provincia	107 Italian Province categorical splits / entity embeddings	9.23%	30.33%	28.22%
• Regione	20 Italian Administrative Regions categorical mapping	6.19%	0.71%	1.26%
• nielsen_area	4 Nielsen Macro-Geographic Areas (NW, NE, Center, South)	3.89%	0.07%	0.45%
• Cap	5-digit Italian Postal Code numerical routing identifier	2.24%	0.79%	0.94%
• Latitudine	Geographic Latitude coordinate (North-South geographic axis)	2.02%	1.16%	1.25%
• Longitudine	Geographic Longitude coordinate (East-West geographic axis)	1.94%	1.54%	1.58%
E. Spatial & Competition	*RVM network proximity, cannibalization, grocery density*	**12.82%**	**6.72%**	**7.33%**
• nearest_coripet_km_log	Distance to nearest operating Coripet RVM (cannibalization proxy)	2.21%	2.68%	2.63%
• is_isolated_rvm	Binary indicator for isolated installation (>5 km from any RVM)	2.27%	0.04%	0.26%
• grocery_supermarket_count	Supermarket retail count in immediate trade catchment	2.16%	1.03%	1.14%
• coripet_rvm_count_5km	Coripet network installation density within 5 km trade radius	2.10%	1.44%	1.51%
• is_local_grocery_monopoly	Binary indicator for sole grocery destination in municipality	2.05%	0.11%	0.30%
• grocery_total_stores	Total commercial food & retail trade presence count	2.03%	1.42%	1.49%
F. Hardware Technology	*RVM machine model, intake capacity, supplier*	**10.22%**	**8.28%**	**8.47%**
• TipoMacchina	RVM hardware model (Ripet_C-front, DAINT_G, Ripet_V2, etc.)	5.89%	7.58%	7.41%
• Fornitore	Hardware manufacturer & supplier (DAINT, RIPET, EURVEN)	4.33%	0.70%	1.06%
G. Retail Brand & Alliance	*Supermarket commercial banner & buying consortium*	**12.61%**	**21.53%**	**20.64%**
• Insegna_Standardized	Commercial retail banner (Conad, Coop, Esselunga, Bennet, etc.)	6.70%	18.40%	17.23%
• nielsen_gruppo	Commercial buying consortium alliance (Conad, Coop, Selex, etc.)	5.91%	3.13%	3.41%
H. Temporal & Seasonality	*Weekly consumer rhythm, holidays, annual hydration cycle*	**13.93%**	**7.76%**	**8.38%**
• is_weekend	Binary flag for Saturday / Sunday peak grocery shopping days	2.37%	0.53%	0.71%
• dow_cos	Cosine day-of-week cyclic harmonic transformation	2.38%	0.20%	0.42%
• is_holiday	Italian national public holiday operating schedule calendar flag	2.06%	2.18%	2.17%
• dow_sin	Sine day-of-week cyclic harmonic transformation	1.89%	2.09%	2.07%
• month_cos	Cosine annual seasonal hydration curve (summer peak cycle)	1.82%	2.44%	2.38%
• month_sin	Sine annual seasonal hydration curve	1.73%	0.26%	0.41%
• quarter	Calendar quarter seasonal progression flag (Q1, Q2, Q3, Q4)	1.67%	0.06%	0.22%

Model Verification Certificate:

• Primary Booster: models/ensemble/lgb_bottiglie.txt (2,500 trees, R^2=0.429) • Neural Network: models/best_model/model.pt (MultiTaskRVMNet PyTorch)

• Production Calibration Blend: 90% LightGBM + 10% MultiTaskRVMNet (models/ensemble/ensemble_config.json)

• All 34 features mathematically audited and verified against preprocessor: models/preprocessing/preprocessor.joblib